

A dynamic early East Antarctic Ice Sheet suggested by ice-covered fjord landscapes

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The first Cenozoic ice sheets initiated in Antarctica from the Gamburtsev Subglacial Mountains¹ and other highlands as a result of rapid global cooling ~34 million years ago². In the subsequent 20 million years, at a time of declining atmospheric carbon dioxide concentrations² and an evolving Antarctic circumpolar current², sedimentary sequence interpretation³ and numerical modelling⁴ suggest that cyclical periods of ice-sheet expansion to the continental margin, followed by retreat to the subglacial highlands, occurred up to thirty times. These fluctuations were paced by orbital changes and were a major influence on global sea levels⁵. Ice-sheet models show that the nature of such oscillations is critically dependent on the pattern and extent of Antarctic topographic lowlands. Here we show that the basal topography of the Aurora Subglacial Basin of East Antarctica, at present overlain by 2–4.5 km of ice, is characterized by a series of well-defined topographic channels within a mountain block landscape. The identification of this fjord landscape, based on new data from ice-penetrating radar, provides an improved understanding of the topography of the Aurora Subglacial Basin and its surroundings, and reveals a complex surface sculpted by a succession of ice-sheet configurations substantially different from today's. At different stages during its fluctuations, the edge of the East Antarctic Ice Sheet lay pinned along the margins of the Aurora Subglacial Basin, the upland boundaries of which are currently above sea level and the deepest parts of which are more than 1 km below sea level. Although the timing of the channel incision remains uncertain, our results suggest that the fjord landscape was carved by at least two ice-flow regimes of different scales and directions, each of which would have over-deepened existing topographic depressions, reversing valley floor slopes.

Deep-sea oxygen isotope records show the onset of significant glaciation in Antarctica at the Eocene/Oligocene boundary^{2,5} (~34 million years (Myr) ago). Morphological evidence for sustained alpine-style glaciation in the Gamburtsev Subglacial Mountains, underlying the Dome A region of the East Antarctic Ice Sheet (EAIS), shows that they were a centre of ice-sheet initiation¹. Although it is thought that the EAIS has remained in a persistent state for the last 14 Myr (as evidenced in the Antarctic Dry Valleys by very low erosion rates⁶, cold-based local glaciers⁷ and the preservation of buried Miocene ice⁸), offshore sedimentary records³ point to there being major oscillations in ice-sheet surface area between 34 and 14 Myr ago. Exactly how these oscillations were expressed by the ice sheet is, however, poorly constrained.

Numerical ice-sheet models can be used to understand the form and flow of past ice sheets. Such models indicate that ice growth begins at higher elevations (such as the Gamburtsev Subglacial Mountains) before encroaching on lower regions^{4,9–11}. The bed elevation grids used as input to these models are, in some regions, constructed from sparse data^{12,13}. One such region is the Aurora Subglacial Basin (ASB; Fig. 1), which from reconnaissance data is known to be a deep trough (more than 1 km below sea level) oriented nearly orthogonal to the modern

ice margin and located to the northeast of the elevated Dome A and Ridge B regions of the ice sheet (Fig. 2a). Ice-sheet models^{10,11} demonstrate the potential importance of the ASB to the progression of ice-sheet growth. These models show a large, growing ice mass from Dome A and Ridge B that converges with smaller radial ice cover from Dome C, resulting in the ASB being buried with deep glacial ice, as at present (Fig. 2c). These models also show that ice-sheet decay is likely to begin in the lowlands of the ASB, isolating a radial ice cap at Dome C and pushing the ice margin back towards Ridge B, eventually depleting the ASB of ice altogether (Fig. 2c). Although it is clear that the ASB has a potentially significant influence on EAIS stability, paucity of bed data, especially around the transition between the ice margin and the interior, is a source of exceptional uncertainty in estimates of the rates and magnitudes of past and present global sea-level changes.

To address this knowledge gap, the ICECAP aerogeophysical programme (Methods) acquired 47,492 line kilometres of airborne radar profiles over the ASB, and from these data a new bed topography has been established (Fig. 2a). The new data extend over a semicircular region radiating from Law Dome and cover approximately $1.5 \times 10^6 \text{ km}^2$ (11% of the Antarctic ice sheet). The region extends from Denman Glacier in

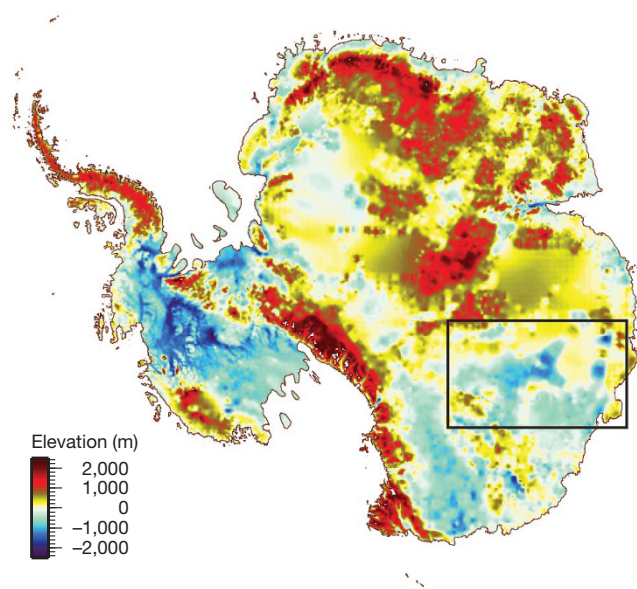


Figure 1 | Bed topography of Antarctica. The blue areas represent Antarctica's major marine subglacial basins. This data set^{12,13} is an interpolation of existing data, which are sparse in the region of the ASB (black box; see also Fig. 2).

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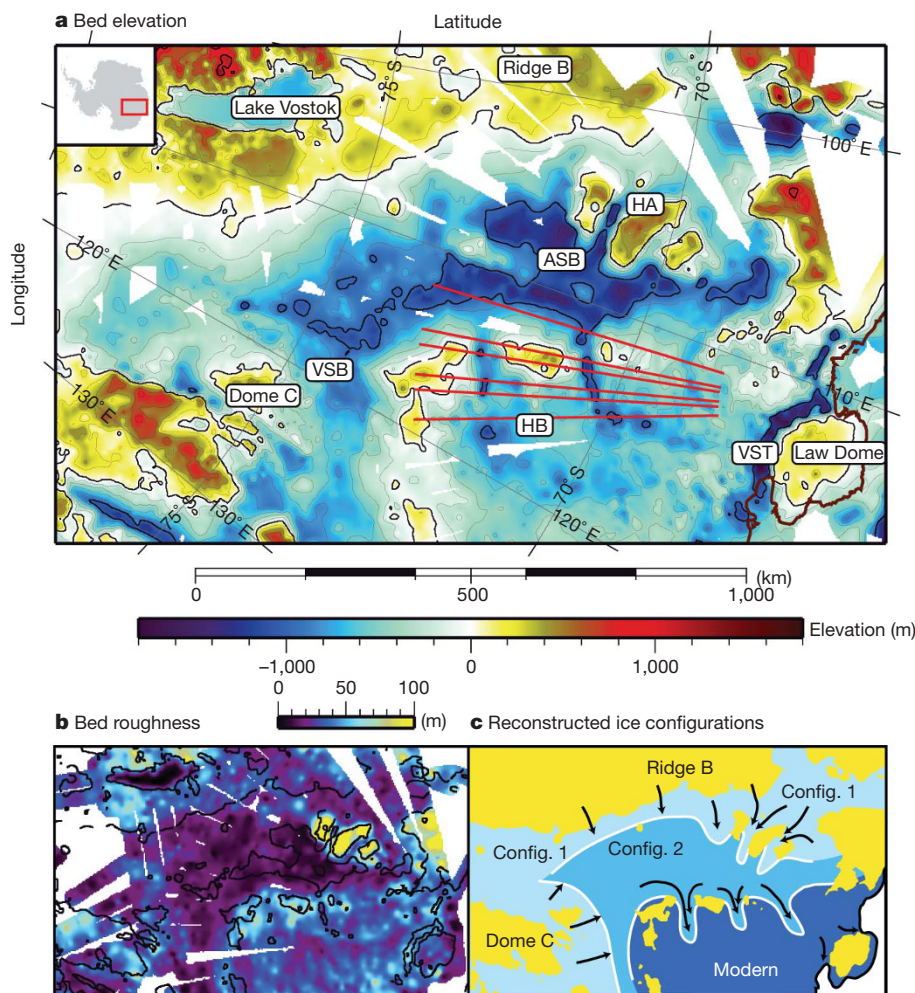


Figure 2 | Bed topography of the ASB region, East Antarctica. **a**, Detailed map derived from radio-echo sounding data from ICECAP, the Support Office for Aerogeophysical Research and BEDMAP, using a natural-neighbour interpolation scheme. Features include subglacial Lake Vostok, Vincennes Subglacial Basin (VSB), Vanderford Subglacial Trench (VST), Law Dome and the ASB. The newly defined features Highland A (HA) and Highland B (HB) are also indicated. See Supplementary Fig. 1 for flight tracks and Supplementary Fig. 2 for data used; regions more than 50 km from data are masked in white.

the west to Dome C in the south and to Porpoise Bay in the east. The deepest point ($-2,426 \pm 10$ m in the WGS-84 coordinate frame) is near the coast in the Vanderford Subglacial Trench, through which both the Vanderford and the Totten Glaciers drain; the highest point ($1,637 \pm 10$ m WGS-84) lies in a previously unknown subglacial mountain range (Highland A) 400 km southeast of Denman Glacier. The thickest ice ($4,522 \pm 10$ m) lies within the trough of the ASB, west of a second subglacial range, Highland B. In an assessment of bed data quality, the average difference in measured ice thicknesses where independently interpreted lines cross was found to be only 33 m.

Southeast portions of the ICECAP map compare well with the gross pattern found in previous compilations (including BEDMAP^{12,13}), which are largely constrained by airborne radar data from collaborative UK-US-Danish surveys from the 1970s¹⁴. Although a new Lagrangian interpolation¹⁵ of the sparse BEDMAP source data incorporating constraints from ice flow shows good general correspondence with the ICECAP data, direct assessment of the ICECAP profile data are required to understand the geomorphology of the region better. We use a conventional natural-neighbour interpolation¹⁶ in this paper.

In the northwest, ICECAP data indicate the presence of a deep depression inland from the Denman Glacier, confirming earlier results; however, instead of the 70,000-km² plateau suggested by the BEDMAP

Red lines are the profiles shown in Fig. 3. Major contours are 1,000 m apart.

b, Gridded along-track root mean squared deviation on an 800-m baseline. Bed elevation contours (500 m) are also marked. Profiles in Supplementary Figs 4 and 5 indicate the morphology of the rough highland regions. **c**, Three stages of ice-sheet development during the glacial fluctuations of the early Miocene or Oligocene epoch. Areas where the bed (without isostatic uplift) is above sea level are shown in yellow. Highland A constrains the edge of configuration 1; Highland B constrains the edge of configuration 2.

compilation, a smaller, intensely dissected mountain region (Highland A) lies to the west of the ASB (also see Supplementary Fig. 4).

The geomorphology uncovered by ICECAP data allows us to infer the nature of the former ice sheets in the region (Fig. 2a). About 20% of the ASB is more than 1 km below sea level, with the deepest regions located between 400 and 700 km inland from the present ice-sheet margin. Along-track bed roughness estimates (Fig. 2b) indicate that the ASB and the adjacent VSB are smooth (low vertical elevation changes over distances less than 1 km) when compared with their surroundings. The smooth domain of the ASB/VSB trough is not purely a function of elevation, but is bounded on its seaward side by a distinct ridge (Highland B) that is ~150 km wide, 0–500 m above sea level and deeply dissected by at least three ~50-km-wide valleys. Between Highland B and the ice-sheet margin, a broad, hummocky region hosts channels emanating from these valleys and heading towards the present-day margin. Highland A, to the west of the ASB, is also deeply dissected by a pair of parallel troughs each of which is more than 50 km wide. Transverse radar profiles reveal continuous troughs that are deepest where they traverse the highland regions (Fig. 3). These troughs reflect a high degree of morphological organization, given their similar size, shape and collinear positioning relative to one another.

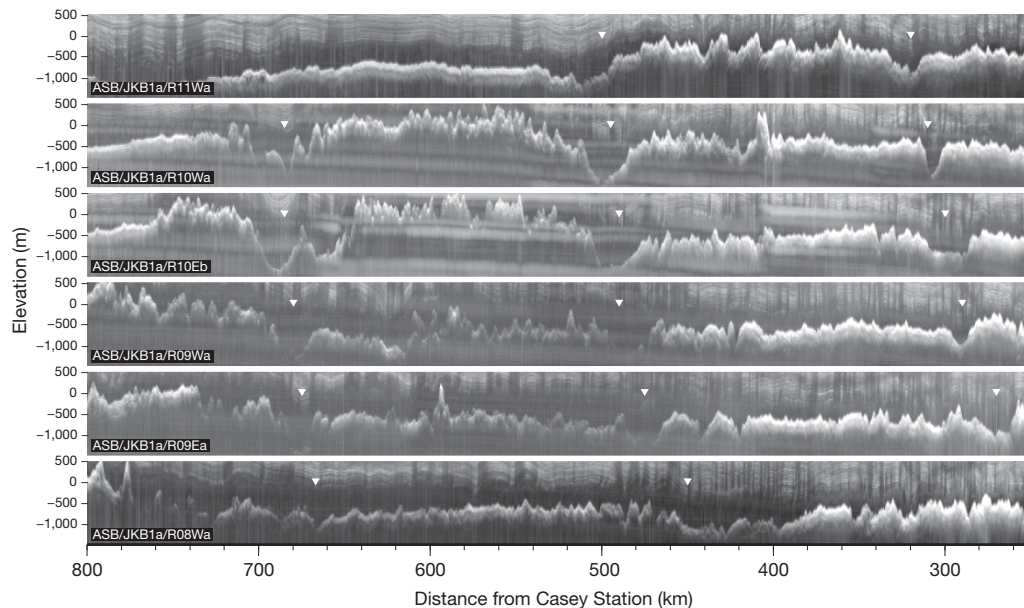


Figure 3 | Bed profiles of fjords in Highland B. Six depth-corrected radar profiles acquired using HiCARS radar are shown. These correspond to the red lines in Fig. 2a, and have similar orientation: south is to the left, west is to the top. Each adjacent profile extends from 500 m above to 1,500 m below sea level and shows a 550-km-long segment. Current sea level is at zero; isostatic uplift models (Supplementary Fig. 6) indicate that the bed may have been more than

500 m higher in the past. The major reflector in each profile is the bed reflection; above that lie layers within the ice. The ice surface is not shown. Fjords show pronounced over-deepening towards the ASB (in the upper half of the figure), which is reached at line R11Wa. Triangles indicate the axes of major through-cutting fjords.

It is clear that the topographic features are likely to be glacial in origin, but the nature of the ice masses responsible for their formation requires discussion. Ice sheets are most erosive near their margins, where high driving stresses, flow velocities and basal pressure gradients combine to produce distinct glacial geomorphology¹⁷. The modern ice sheet is unlikely to be capable of developing these features, because the regional ice flow is slow and cuts across trough axes (Supplementary Fig. 3). Hence, the morphology relates to past glaciations and to ice-sheet configurations different from those found today. Deep troughs selectively breaching uplands near the margins of ice sheets are well known in the Northern Hemisphere and occur in western Norway, East Greenland and eastern Baffin Island^{18–20}. Often such troughs have deepened pre-existing river valleys^{18–20}. In situations where such fjord troughs cut through the main upland axis, there is a close correlation between the depth of the trough and the height of the constraining uplands. Selective ice flow near an ice margin is favoured by two factors. First, the mountain barrier has a proportionately larger impact where the ice is thin and, second, ice velocities increase towards the ice-sheet margin. Under such conditions, low points in the topography are deepened and the more they deepen the more ice they drain^{21,22}, until an erosion threshold is reached¹⁷. In Antarctica, similar landscapes, such as the Transantarctic Mountains and the mountain front parallel to the coast in Dronning Maud Land, are associated with mountains acting as a barrier to ice flow and bounding the ice sheet.

Our understanding of the relationship between ice sheets and subglacial topography has benefited from analyses of formerly glaciated terrain and its glacial history, especially in Scandinavia. Here numerous episodes of glaciation centred on the main upland axis between Norway and Sweden have left a divergent pattern of regional troughs, and these have been overprinted by a continental-scale flow of ice across the region²³.

Two glacial configurations, in addition to the modern ice-sheet form, are inferred for the ASB. Configuration 1 involves ice flow from Ridge B that cuts deep channels in at least two places into Highland A to the west of the ASB. No ice from the Dome C region is needed to cut these valleys. A second series of valleys initiating in Highland B demonstrates ice flowing fast through the uplands east of the ASB

and across the broad, flat region towards the present-day margin. From this, we infer configuration 2, which is significantly larger in area than configuration 1 and involves convergent flow from Dome C and Ridge B into the ASB (Fig. 2c). The over-deepened troughs are formed through convergence of fast-flowing ice. They are manifest as topography shallowing downstream with reverse bed slope in the direction of the ice flow. Their formation requires an environment with abundant subglacial water¹⁷, probably requiring significant surface melt, analogous to Quaternary Northern Hemisphere glaciations (which are also noted for their oscillatory nature). The smooth landscape of the ASB upstream of fjords in configuration 2 is typical of a regime of enhanced erosion and deposition.

As in Scandinavia, we expect the valleys and troughs to show reactivation over time rather than to depict a single glacial event. In this way, the ASB may have experienced numerous glacial advances and recessions, many of which will have been orbitally paced. The glaciological reconstructions we infer from the landscape are consistent with the numerical models of growth and decay^{4,9,11}, despite the lack of detailed bed information in this sector informing these models. Ice growth and decay across the portions of the ASB that lie above sea level probably requires surface melt not currently present in Antarctica and, hence, temperatures significantly higher than at present. Such conditions, and therefore such ice sheets, have been restricted to the Northern Hemisphere over the past 14 Myr; hence, the formation of the landforms identified in the ASB highlands and, most probably, elsewhere in East Antarctica probably dates from the early Miocene or Oligocene epoch. If this is the case, the large oscillations in ice volume, paced by orbital changes, observed in offshore sequences²⁴ can be explained. An alternative view, that the glacial landforms were formed in the Pliocene epoch²⁴, requires the loss of much of the Antarctic ice sheet, with implications for global temperatures and sea levels.

Although it is difficult to know with certainty the topographic elevation of the ASB region during early EAIS oscillations, if the present ice sheet were removed the region seawards of the ice-cut fjords would be around sea level after isostatic uplift, whereas the ASB itself would remain substantially below sea level (Supplementary Fig. 6). As ice sheets are known to be sensitive to environmental change in such

lowland and shallow marine settings²⁵, we are able to infer the likely glacial processes responsible for changes in former ice sheets. Ice-sheet retreat to configuration 1 from configuration 2 may involve a marine instability similar to that proposed as being relevant to West Antarctica²⁶, in which ice retreat is associated with water depth increase at the margin and enhanced loss of ice through calving and melting leads to deglaciation of the entire ASB. Growth from configuration 1 to configuration 2 is more difficult to achieve, as it requires a major deep basin, filled with water, to be filled by grounded ice. Some have argued that deep, pre-glacial lakes such as Lake Vostok may have survived glaciation as subglacial lakes²⁷. Others have recognized the absence of large subglacial lakes in some troughs as evidence for migration of a grounded margin during ice growth²⁸. This is because a steep marginal surface gradient would drive water to the edge of the ice sheet. The absence of a large subglacial lake within the interior basin points to the latter explanation for its glaciation.

Evidence of fjords in East Antarctica cut by ice sheets of varying configuration may not be limited to our study region. Measurement of comparable features may allow us to appreciate better the magnitude of early EAIS change and the processes responsible.

METHODS SUMMARY

We used a ski-equipped, long-range DC-3T carrying a HiCARS coherent, 60-MHz, ice-penetrating radar²⁹ along with a gravimeter, magnetometers and laser altimeters. The out-and-back aircraft survey range is ~1,000 km. Twenty-six flights were supported by Casey Station in December–January of 2008–2009 and 2009–2010. Radial flights from Casey Station were undertaken to maximize coverage of the interior, along with reflights of ICESAT orbital tracks and coast-parallel tie lines. Radar data were pulse-compressed and processed using a short synthetic-aperture radar aperture to retain energy; with this level of processing, range distortions are not significant on length scales greater than 400 m. The ice thickness was found using a speed of light in ice of $169 \text{ m } \mu\text{s}^{-1}$, and the bed elevation was calculated using the radar-determined surface elevation. These new data were combined with data from BEDMAP and the Support Office for Aerogeophysical Research, and interpolated using a natural-neighbour algorithm¹⁶. Such algorithms are commonly used with irregularly distributed data confined to discrete transects. We determined along-track roughness using the root mean squared deviation³⁰ of detrended bed elevation data on an 800-m baseline.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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